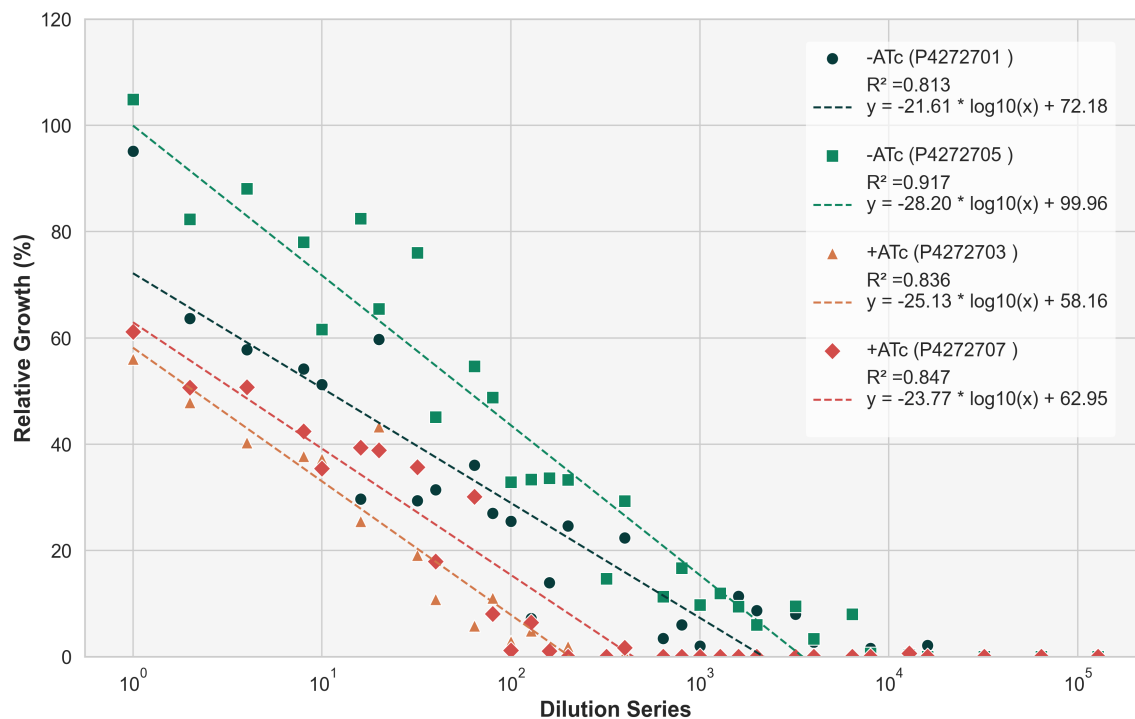
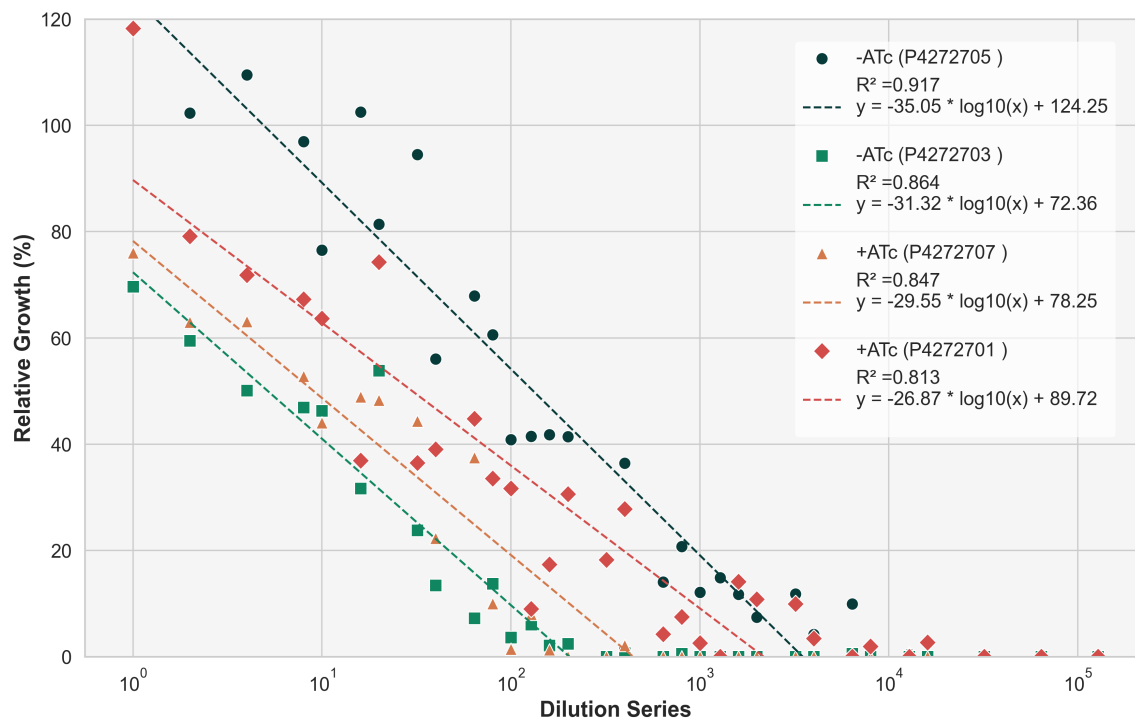


Growth Curve for E. coli K-12

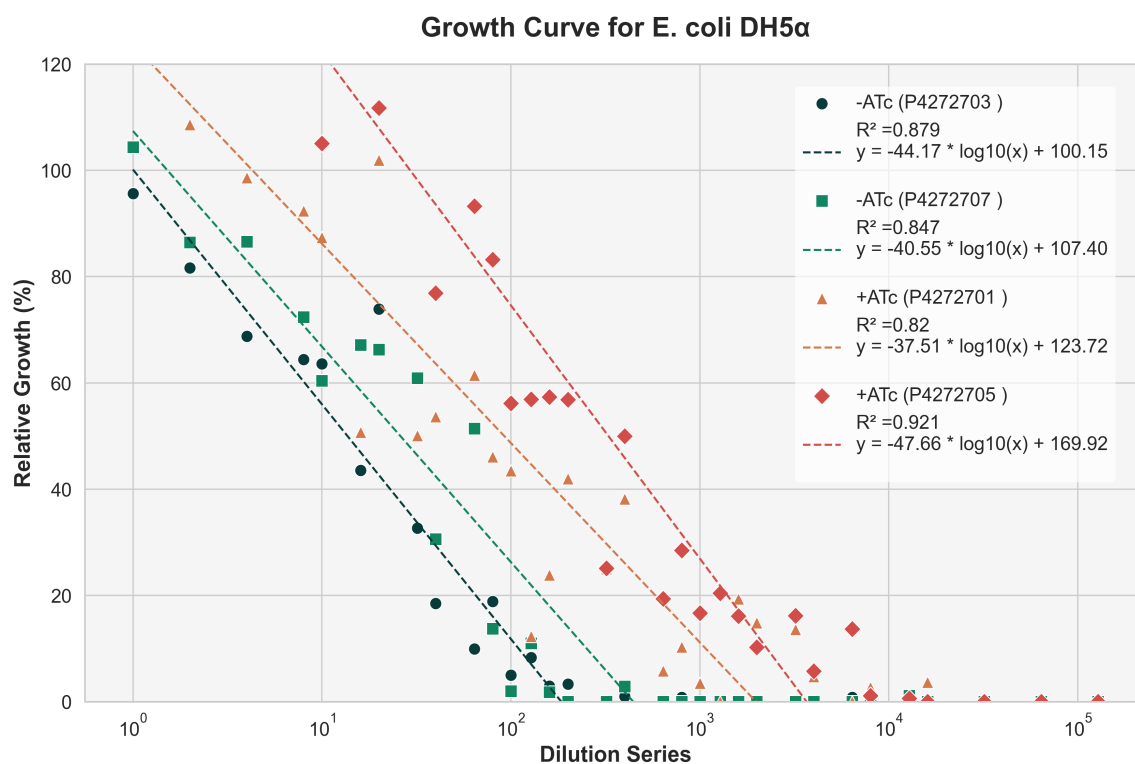


	P4272701	P4272705	P4272703	P4272707
Formula	$y = -21.61 * \log_{10}(x) + 72.18$	$y = -28.20 * \log_{10}(x) + 99.96$	$y = -25.13 * \log_{10}(x) + 58.16$	$y = -23.77 * \log_{10}(x) + 62.95$
Slope	-21.61	-28.20	-25.13	-23.77
Intercept	72.18	99.96	58.16	62.95
R-squared	0.813	0.917	0.836	0.847
Y-cut	72.18	99.96	58.16	62.95
X-cut	2184.03	3508.84	206.18	444.99
X at Y=50	10.62	59.14	2.11	3.51

Growth Curve for E. coli BL21



	P4272705	P4272703	P4272707	P4272701
Formula	y = -35.05 * log ₁₀ (x) + 124.25	y = -31.32 * log ₁₀ (x) + 72.36	y = -29.55 * log ₁₀ (x) + 78.25	y = -26.87 * log ₁₀ (x) + 89.72
Slope	-35.05	-31.32	-29.55	-26.87
Intercept	124.25	72.36	78.25	89.72
R-squared	0.917	0.864	0.847	0.813
Y-cut	124.25	72.36	78.25	89.72
X-cut	3508.84	204.39	444.99	2184.03
X at Y=50	131.39	5.17	9.04	30.08



	P4272703	P4272707	P4272701	P4272705
Formula	y = -44.17 * log ₁₀ (x) + 100.15	y = -40.55 * log ₁₀ (x) + 107.40	y = -37.51 * log ₁₀ (x) + 123.72	y = -47.66 * log ₁₀ (x) + 169.92
Slope	-44.17	-40.55	-37.51	-47.66
Intercept	100.15	107.40	123.72	169.92
R-squared	0.879	0.847	0.820	0.921
Y-cut	100.15	107.40	123.72	169.92
X-cut	185.13	444.99	1986.48	3677.37
X at Y=50	13.66	26.03	92.29	328.36

Filename: test_image.jpg
Type: Growth Curve
Detergent: SDS
Treatment: Heat Shock
Repeat: 1
Strain A: E. coli K-12
Strain B: E. coli BL21
Strain C: E. coli DH5 α
Parameters Used:
Threshold: 128
Minimum Area: 50



Contours Image



Binary Image



Grid Image



This report was generated by **Spotplotter** version **1.0** on **2024-10-09**

Spotplotter was created by **Holly Lewis** with supervision from **R Verrinder** and **Dr. M Mason** as BSc (Eng) final year project submitted in partial fulfilment of the requirements for the degree of Bachelor of Science in Electrical and Computer Engineering in the Department of Electrical Engineering at the University of Cape Town.

To read the full report please see: *GITHUB LINK*

Formula: The formula represents the linear regression equation that models determined using the `linregress` function from `scipy.stats` which determines the relationship between the logarithm of the dilution series and relative growth. It follows the form: $y = m \cdot \log_{10}(x) + b$

Normalization: The relative growth values were normalized to a baseline to make the results comparable across different conditions. The quantified values of each spot were divided by the average value of the first spot in the -ATP for each strain.

The slope (m): indicates the rate of change in relative growth as the dilution series increases (on a logarithmic scale). A steep slope indicated that the growth has a faster knockdown as the dilution changes. A shallow slope indicates that the growth is more stable across dilutions.

The intercept (b) and y-cut is the relative growth when the solution is not diluted.

The R-squared value measures how well the linear regression line fits the data, where 1 represents a perfect fit and 0 represents no relationship. Higher R^2 values indicates that the growth follows a linear relationship.

The X-cut refers to the point where the regression line crosses the x-axis, indicating the dilution value at which the relative growth would theoretically be zero (no growth).

X at Y = 50: This value represents the dilution series value when the relative growth is 50% i.e., the knockdown is 50% in comparison to the -ATP series.